

Bangladesh University of Business and Technology



LAB REPORT

Experiment No : 03
Course Title : Operating Systems Lab
Course Code : CSE 210
Experiment Name : Basic Shell Scripting Using Conditional Statements and File Operations

Submitted By:

Name : Luckman Rafique
ID No : 20255103042
Intake : 55
Section : 02
Program : B.Sc. in CSE

Submitted To:

Name : Mahbuba Habib

(Lecturer)
Dept. of CSE
**Bangladesh University of Business
& Technology**

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QUESTION 01 - Take two numbers as input from the user. Perform an arithmetic operation (x+y-z) on three numbers using Shell Script.

Steps To Solve –

- Open “Git Bash”
- Go inside the Operating System folder
 - cd “Operating System”
- Then go inside Lab Task folder
 - cd “Lab Task”
- Create a folder named Lab 3
 - mkdir “Lab 3”
- Create a shell script named q1.sh
 - touch q1.sh
- Open the q1.sh using vi.
 - vi q1.sh
- Write the required Shell Script code and save the file.
- Give execute permission to the file.
 - chmod u+x q1.sh
- Run the Shell Script.
 - ./q1.sh
- Enter the first, second, and third numbers.
- The program performs the arithmetic operation (first number + second number - third number) and displays the result.

Code –

```
echo Enter first number:
read x
echo Enter second number:
read y
echo Enter third number:
read z

echo Result: $(((x+y)-z))
```

Output –

```
User@DESKTOP-F308A5A MINGW64 /e/#/Journey-Towards-Growth/Semester 5 (main)
$ cd "Operating System"

User@DESKTOP-F308A5A MINGW64 /e/#/Journey-Towards-Growth/Semester 5/Operating System (main)
$ ls
'Lab Report'/'Lab Task'/

User@DESKTOP-F308A5A MINGW64 /e/#/Journey-Towards-Growth/Semester 5/Operating System (main)
$ cd "Lab Task"

User@DESKTOP-F308A5A MINGW64 /e/#/Journey-Towards-Growth/Semester 5/Operating System/Lab Task (main)
$ mkdir "Lab 3"

User@DESKTOP-F308A5A MINGW64 /e/#/Journey-Towards-Growth/Semester 5/Operating System/Lab Task/Lab 3 (main)
$ touch q1.sh

User@DESKTOP-F308A5A MINGW64 /e/#/Journey-Towards-Growth/Semester 5/Operating System/Lab Task/Lab 3 (main)
$ vi q1.sh

User@DESKTOP-F308A5A MINGW64 /e/#/Journey-Towards-Growth/Semester 5/Operating System/Lab Task/Lab 3 (main)
$ chmod u+x q1.sh

User@DESKTOP-F308A5A MINGW64 /e/#/Journey-Towards-Growth/Semester 5/Operating System/Lab Task/Lab 3 (main)
$ ./q1.sh
Enter first number:
5
Enter second number:
5
Enter third number:
2
Result: 8
```

QUESTION 02 - Take marks as input from the user and display a message according to the marks.

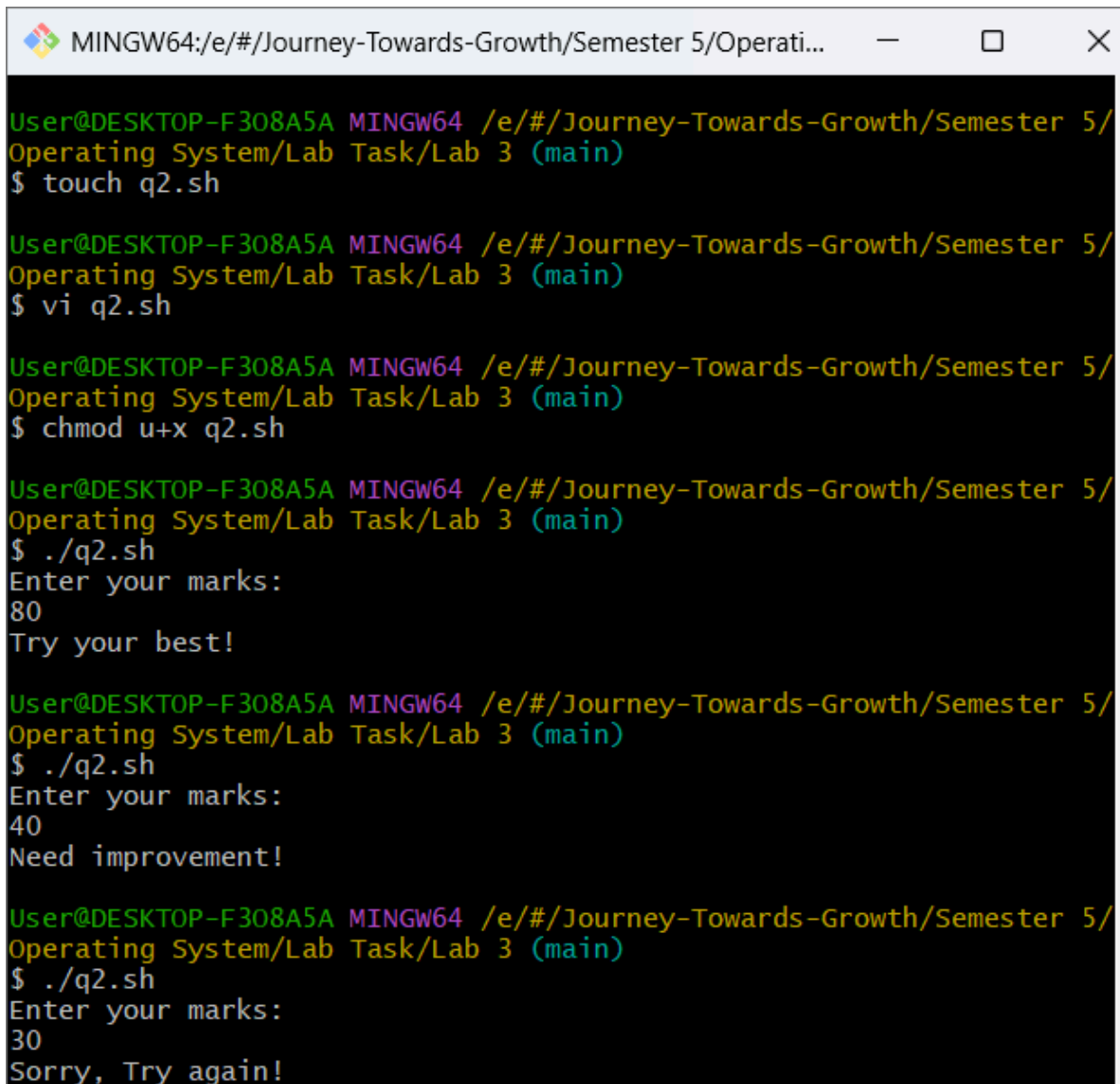
Steps To Solve –

- Create a shell script named q2.sh
 - touch q2.sh
- Open the q2.sh using vi.
 - vi q2.sh
- Write the required Shell Script code and save the file.
- Give execute permission to the file.
 - chmod u+x q2.sh
- Run the Shell Script.
 - ./q2.sh
- Enter the marks.
- The program checks the marks using conditional statements.
- If the marks are above 80, it displays “Good, Keep it up!”
- If the marks are above 60, it displays “Try your best!”
- If the marks are 40 or above, it displays “Need improvement!”
- Otherwise, it displays “Sorry, Try again!”

Code –

```
echo Enter your marks:
read a
if [ $a -gt 80 ] ; then
    echo Good, Keep it up!
elif [ $a -gt 60 ] ; then
    echo Try your best!
elif [ $a -ge 40 ] ; then
    echo Need improvement!
else
    echo Sorry, Try again!
fi
```

Output –



```
MINGW64:/e/#!/Journey-Towards-Growth/Semester 5/Operati...
User@DESKTOP-F308A5A MINGW64 /e/#!/Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ touch q2.sh

User@DESKTOP-F308A5A MINGW64 /e/#!/Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ vi q2.sh

User@DESKTOP-F308A5A MINGW64 /e/#!/Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ chmod u+x q2.sh

User@DESKTOP-F308A5A MINGW64 /e/#!/Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ ./q2.sh
Enter your marks:
80
Try your best!

User@DESKTOP-F308A5A MINGW64 /e/#!/Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ ./q2.sh
Enter your marks:
40
Need improvement!

User@DESKTOP-F308A5A MINGW64 /e/#!/Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ ./q2.sh
Enter your marks:
30
Sorry, Try again!
```

QUESTION 03 - Take two numbers as input and perform addition or subtraction based on their values.

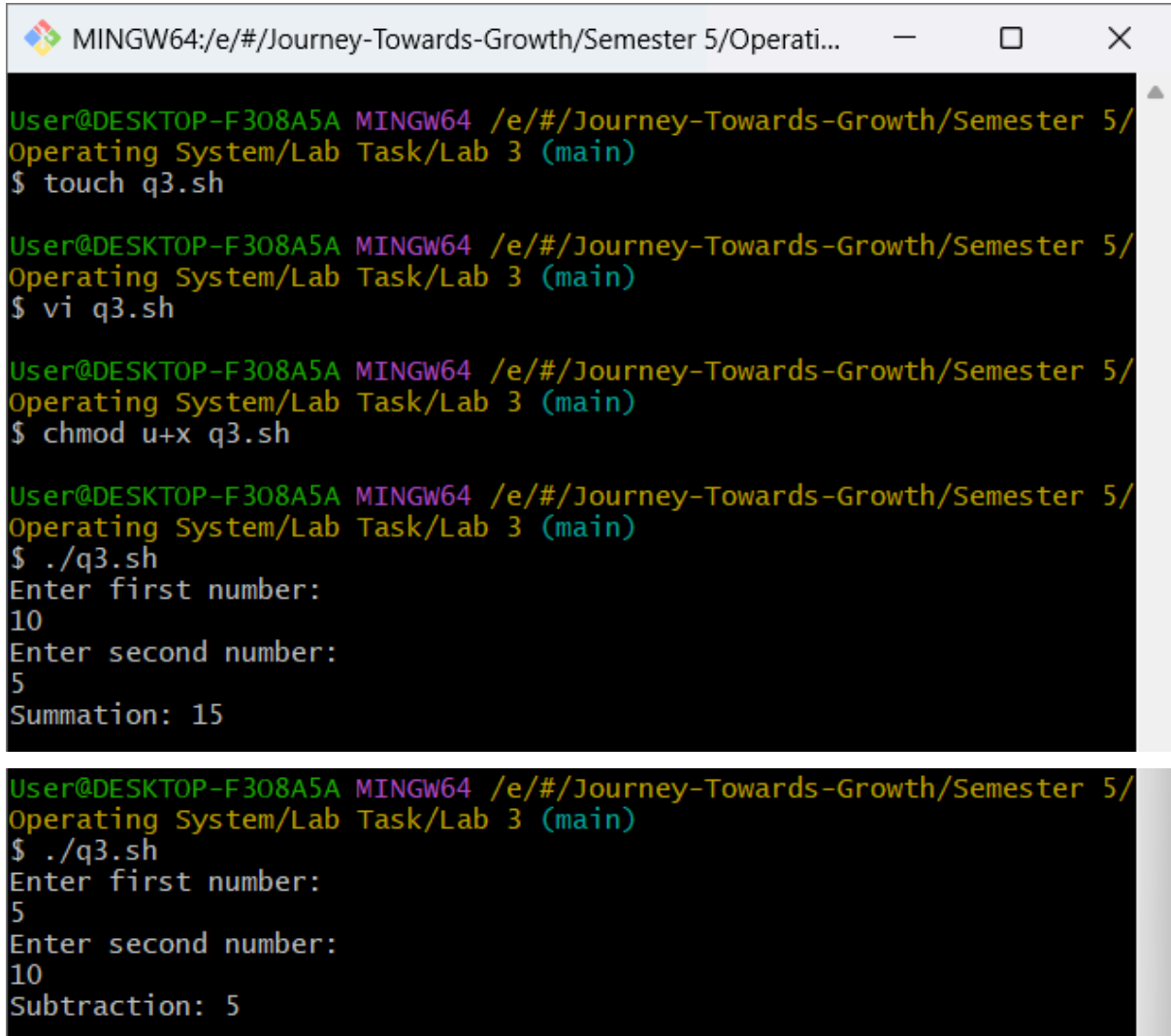
Steps To Solve –

- Create a shell script named q3.sh
 - touch q3.sh
- Open the q3.sh using vi.
 - vi q3.sh
- Write the required Shell Script code and save the file.
- Give execute permission to the file.
 - chmod u+x q3.sh
- Run the Shell Script.
 - ./q3.sh
- Enter the first and second numbers.
- The program compares the two numbers.
- If the first number is greater than the second number, it calculates their addition.
- Otherwise, it calculates the subtraction of the first number from the second number.
- The result is displayed on the screen.

Code –

```
echo Enter first number:
read a
echo Enter second number:
read b
if [ $a -gt - $b ] ; then
    echo Summation: $((a+b))
else echo Subtraction: $((b-a))
fi
```

Output –



```
MINGW64:/e/##Journey-Towards-Growth/Semester 5/Operati...
User@DESKTOP-F308A5A MINGW64 /e/##Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ touch q3.sh

User@DESKTOP-F308A5A MINGW64 /e/##Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ vi q3.sh

User@DESKTOP-F308A5A MINGW64 /e/##Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ chmod u+x q3.sh

User@DESKTOP-F308A5A MINGW64 /e/##Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ ./q3.sh
Enter first number:
10
Enter second number:
5
Summation: 15

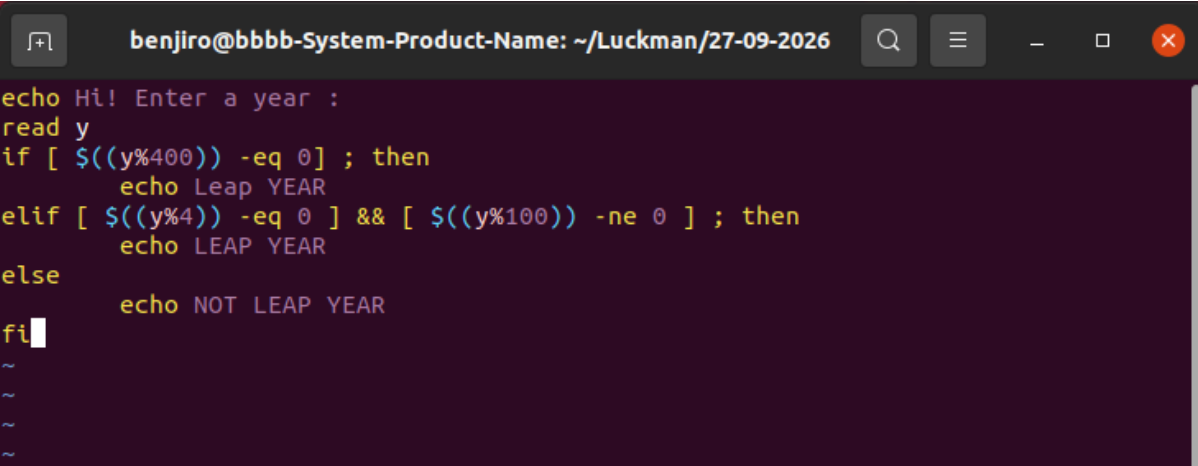
User@DESKTOP-F308A5A MINGW64 /e/##Journey-Towards-Growth/Semester 5/
Operating System/Lab Task/Lab 3 (main)
$ ./q3.sh
Enter first number:
5
Enter second number:
10
Subtraction: 5
```

QUESTION 04 - Take a year as input and check whether it is a leap year or not.

Steps To Solve –

- Create a shell script named q4.sh
 - touch q4.sh
- Open the q4.sh using vi.
 - vi q4.sh
- Write the required Shell Script code and save the file.
- Give execute permission to the file.
 - chmod u+x q4.sh
- Run the Shell Script.
 - ./q4.sh
- Enter a year.
- The program checks whether the year is divisible by 400.
- If not, it checks whether the year is divisible by 4 but not by 100.
- If either condition is true, it displays “Leap Year.”
- Otherwise, it displays “Not Leap Year.”

Code –



```
benjiro@bbbb-System-Product-Name: ~/Luckman/27-09-2026
echo Hi! Enter a year :
read y
if [ $((y%400)) -eq 0 ] ; then
    echo Leap YEAR
elif [ $((y%4)) -eq 0 ] && [ $((y%100)) -ne 0 ] ; then
    echo LEAP YEAR
else
    echo NOT LEAP YEAR
fi
```

Output –

```
benjiro@bbbb-System-Product-Name: ~/Luckman/27-09-2026
LEAP YEAR
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ vi t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ chmod u+x t.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ ./t.sh
Hi! Enter a year :
2020
LEAP YEAR
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ ./t.sh
Hi! Enter a year :
2023
NOT LEAP YEAR
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ ./t.sh
Hi! Enter a year :
2025
NOT LEAP YEAR
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ ./t.sh
Hi! Enter a year :
2026
NOT LEAP YEAR
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ ./t.sh
Hi! Enter a year :
1999
NOT LEAP YEAR
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$
```

QUESTION 05 - Create a folder named Test-2, create a Shell Script named ifelse.sh to take three numbers as input and find the lowest number, copy ifelse.sh to sad.sh, run the program through sad.sh, and finally rename sad.sh to happy.sh.

Steps To Solve –

- **Create a folder named Test-2.**
 - **mkdir Test-2**
- **Go inside the Test-2 folder.**
 - **cd Test-2**
- **Create a shell script named ifelse.sh**
 - **touch ifelse.sh**
- **Open the ifelse.sh file using vi.**
 - **vi ifelse.sh**
- **Write the required Shell Script code and save the file.**
- **Give execute permission to the file.**
 - **chmod u+x ifelse.sh**
- **Run the Shell Script.**
 - **./ifelse.sh**
- **Enter three numbers as input.**
- **The program compares the three numbers and finds the lowest number.**
- **The lowest value is displayed as output.**
- **Copy ifelse.sh to sad.sh**
 - **cp ifelse.sh sad.sh**
- **Run the program through sad.sh.**
 - **./sad.sh**
- **Finally, rename sad.sh to happy.sh.**
 - **mv sad.sh happy.sh**
- **Run the program again through happy.sh.**
 - **./happy.sh**

Code –

```
echo Enter first number:
read a
echo Enter second number:
read b
echo Enter third number:
read c
if [ $a -lt $b ] && [ $a -lt $c ] ; then
    echo The lowest value is: $a
elif [ $b -lt $a ] && [ $b -lt $c ] ; then
    echo The lowest value is: $b
else echo The lowest value is: $c
fi
~
```

Output –

```
benjiro@
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ mkdir Test2
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ ls
hello.sh  small.sh  Test2  test.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026$ cd Test2
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ touch ifelse.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ vi ifelse.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ chmod ifelse.sh
chmod: missing operand after 'ifelse.sh'
Try 'chmod --help' for more information.
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ chmod u+x ifelse.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ ./ifelse.sh
Enter first number:
5
Enter second number:
2
Enter third number:
6
./ifelse.sh: line 12: syntax error: unexpected end of file
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ vi ifelse.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ chmod u+x ifelse.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ ./ifelse.sh
Enter first number:
5
Enter second number:
6
Enter third number:
8
The lowest value is: 5
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ ./ifelse.sh
Enter first number:
5
Enter second number:
2
Enter third number:
3
The lowest value is: 2
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ cp ifelse.sh sad.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ ls
ifelse.sh  sad.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ chmod u+x sad.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ ./sad.sh
Enter first number:
5
Enter second number:
2
Enter third number:
6
The lowest value is: 2
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ mv sad.sh happy.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$ ls
happy.sh  ifelse.sh
benjiro@bbbb-System-Product-Name:~/Luckman/27-09-2026/Test2$
```